Chapter 1. Abstract Integration.

1. Does there exist an infinite σ-algebra which has only countably many members? (Answer: No!)

Proof. Let $\mathcal{M}$ be a σ-algebra on a set X.
Suppose that $\mathcal{M}$ has only countably infinite members.
If X is finite set, then the power set $\mathcal{P}(X)$ is finite, so is $\mathcal{M} \subset \mathcal{P}(X)$.
Therefore, X must be infinite if $\mathcal{M}$ is infinite.
Since $\mathcal{M}$ is countably infinite, enumerate $\mathcal{M} = \{A_i \mid i \in \mathbb{N}\}$.
For each $x \in X$, define $B_x := \bigcap_{\substack{x \in A, \\ A \in \mathcal{M}}} A$. Then, B_x is a countable intersection of $A_i \in \mathcal{M}$, so $B_x \in \mathcal{M}$.

Assume that: for any $x, y \in X$, $B_x \cap B_y \neq \emptyset$. Let $z \in B_x \cap B_y$.
Then, for all $A \in \mathcal{M}$ containing x, $z \in A$. So,

$$B_z = \bigcap \{A \mid A \in \mathcal{M},\ z \in A\} \subseteq \bigcap \{A' \mid A' \in \mathcal{M},\ x \in A'\} = B_x$$

and similarly $B_z \subseteq B_y$, thus $z \in B_z \subseteq B_x \cap B_y$.
Meanwhile, assume that $x \notin B_z$. Then, $x \in B_x \setminus B_z$. But, since $B_x, B_z \in \mathcal{M}$, $B_x \setminus B_z = B_x \cap B_z^c \in \mathcal{M}$, so that $B_x \setminus B_z = B_x$. It contradicts to $z \in B_x$.
So, $x \in B_z$ and similarly $y \in B_z$. Therefore $B_z = B_x \cap B_y = B_x = B_y$.
Let $\mathcal{B} := \{B_x \mid x \in X\}$. Assume $\mathcal{B}$ is finite. Notice that: given $A \in \mathcal{M}$,

$$A = \bigcup_{x \in A} B_x, \text{ so that } \quad \#\mathcal{M} \leq \#\{0,1\}^{\mathcal{B}} < \aleph_0,$$

but it is contradition, since $\#\mathcal{M} = \aleph_0$.
Assume $\mathcal{B}$ is uncountably infinite. Then, since $\mathcal{B} \subseteq \mathcal{M}$, $\mathcal{M}$ is also uncountable, so it contradicts.
Now, suppose $\mathcal{B}$ is countable, and let enumerate $\mathcal{B} = \{B_i \mid i \in \mathbb{N}\}$ which satisfies that $B_i = B_j \Leftrightarrow i = j$ and $i \neq j \Rightarrow B_i \cap B_j$.
Then, for any $F_1, F_2 \subseteq \mathbb{N}$, $F_1 \neq F_2$ implies that

$$\bigcup_{i \in F_1} B_i \neq \bigcup_{j \in F_2} B_j, \text{ and } \bigcup_{n \in F} B_n \in \mathcal{M} \text{ for each } F \subseteq \mathbb{N}.$$

Now, a map

$$\mathbb{N} \to \mathcal{M} : F \mapsto \bigcup_{n \in F} B_n$$

is well-defined injective map.
Therefore, $\#\mathcal{P}(\mathbb{N}) = 2^{\aleph_0} = \#\mathcal{M}$, it contradicts, so $\mathcal{M}$ is uncountable. ∎

Similar question
Does there exists an infinite topology which has only countably many members?

Yes. Let $X=\mathbb{N}$ with cofinite topology.
But, there is a fun fact:

Infinite Hausdorff topology must be uncountable.

Proof. Let $(X,\mathcal{T})$ be a Hausdorff space, the $\mathcal{T}$ is infinite set.
Since $\mathcal{T}\subseteq\mathcal{P}(X)$, X must be infinite. Suppose that $\mathcal{T}$ is countable.
Assume that X has infitely many isolated points, denote $\{x_n \mid n\in\mathbb{N}\}$.
Then, a map $\{S \mid S\subseteq\mathbb{N}\}\to\mathcal{T} : S\mapsto \bigcup_{n\in S}\{x_n\}$ is well-defined injective map.
Since $|\mathcal{P}(\mathbb{N})| = 2^{\aleph_0}$, so that $|\mathcal{T}|\geq 2^{\aleph_0}$.
Now, assume that X has only finitely many isolated points.
Since X is infinite, there exists a non-isolated point $p\in X$.
Put $U_0=X$. Since p is non-isolated, choose $x_0\in U_0\setminus\{p\}$.
By the T_2 condition, there are disjoint open sets U_1 and V_0 s.t

$$p\in U_1,\ x_0\in V_0 \text{ and } U_1\cap V_0=\emptyset.$$

Now, for $n\geq 1$, choose $x_n\in U_n\setminus\{p\}$. Then, by the T_2 again,
there are disjoint open sets U' and V' such that

$$p\in U' \text{ and } x_n\in V', \text{ and } U'\cap V'=\emptyset.$$

Put $U_{n+1}=U_n\cap U'$, $V_n=U_n\cap V'$. Now, $p\in U_n$, $x_n\in V_n$ for all $n\in\mathbb{N}_0$,
and $\{V_n \mid n\in\mathbb{N}\}$ is pairwise disjoint nonempty open sets. Now,

$$\mathcal{P}(\mathbb{N})\to\mathcal{T} : A\mapsto \bigcup_{n\in A} V_n \text{ is injective, so proves.}$$ □

Lemma. $\mathcal{P}(\mathbb{N})$ is uncountable.
Proof. Assume that $\mathcal{P}(\mathbb{N})$ is countable. Enumerate $\mathcal{P}(\mathbb{N})=\{A_n \mid n\in\mathbb{N}\}$.
Define $B=\{n\in\mathbb{N} \mid n\notin A_n\}$. Then, B is a subset of $\mathbb{N}$, thus

$$B=A_k \text{ for some } k\in\mathbb{N}.$$

But, $k\in B \Leftrightarrow k\notin A_k \Leftrightarrow k\notin B$, so it contradicts to the hypothesis. □

3. Prove that: if f is a real function on a measurable space X such that $\{x \in X \mid f(x) \geq r\}$ is measurable for all $r \in \mathbb{Q}$, then f is measurable.

Proof. Let $(a,b) \subseteq \mathbb{R}$ be an open interval with $a<b$ and $a,b \in \mathbb{Q}$.

$$f^{-1}[(a,b)] = f^{-1}[(-\infty, b)] \cap f^{-1}[(a,\infty)]$$

$$= \left(f^{-1}[[b,\infty)]\right)^c \cap \left(\bigcup_{\substack{r \in \mathbb{Q} \\ r>a}} f^{-1}[[r,\infty)]\right), \text{ so proved. } \square$$

4. Let $\{a_n\}$ and $\{b_n\}$ be sequences in $[-\infty, \infty]$. Prove that

$$\limsup_{n\to\infty}(-a_n) = -\liminf_{n\to\infty} a_n$$

and

$$\limsup_{n\to\infty}(a_n + b_n) \le \limsup_{n\to\infty} a_n + \limsup_{n\to\infty} b_n.$$

and if $a_n \le b_n$ for all $n \in \mathbb{N}$, $\liminf_{n\to\infty} a_n \le \liminf_{n\to\infty} b_n$.

5.a) Suppose that $f, g: X \to [-\infty, \infty]$ are measurable.
Prove that the sets $\{x \in X \mid f(x) < g(x)\}$, $\{x \in X \mid f(x) = g(x)\}$

are measurable.
Proof. Since f, g are measurable, so $f - g$ is measurable.
And, notice that
$\{x \in X \mid f(x) < g(x)\} = (f-g)^{-1}[[-\infty, 0)]$ and $\{x \in X \mid f(x) = g(x)\} = (f-g)^{-1}[\{0\}]$
So both are measurable.

5.b) Let $f_n: X \to \mathbb{R}$ be a sequence of real-valued functions.
Prove that the set $\{x \in X \mid \{f_n(x)\} \text{ converges}\}$

is measurable set.

6. Let X be an uncountable set, and let

$$\mathcal{M} \overset{\text{def}}{=} \{E \subseteq X \mid \text{either } E \text{ or } E^c \text{ is at most countable}\}.$$

$$= \{E \subseteq X \mid |E| \leq |\mathbb{N}|\} \cup \{E \subseteq X \mid |E^c| \leq |\mathbb{N}|\}.$$

Define $\mu : \mathcal{M} \to [0, \infty] : E \mapsto \begin{cases} 0 & \text{if } |E| \leq |\mathbb{N}| \\ 1 & \text{if } |E^c| \leq |\mathbb{N}|. \end{cases}$

Prove that $\mathcal{M}$ is a σ-algebra and μ is a measure on $\mathcal{M}$. Describe the corresponding measurable functions and thier integral.

Proof. Notice that, since $X = E \cup E^c$, the case that both E and E^c are at most countable cannot be happen. Therefore, μ is well-defined.

First, clearly $\emptyset, X \in \mathcal{M}$, because $X^c = \emptyset$.

Let $E_n \in \mathcal{M}$ $(n \in \mathbb{N})$ be given. Suppose $\bigcup_{n=1}^{\infty} E_n$ is uncountable. Then, since countable union of countable sets is countable, so that at least one of E_n is uncountable, say E_k. Now,

$$\left(\bigcup_{n=1}^{\infty} E_n\right)^c = \left(\bigcap_{n=1}^{\infty} E_n^c\right) \subseteq E_k^c, \text{ thus } \bigcup_{n=1}^{\infty} E_n \in \mathcal{M}.$$

And, $\mathcal{M}$ is closed under the complement by definition, so $\mathcal{M}$ is σ-algebra.

Now, to show μ satisfies countable subadditivity, let $E_n \in \mathcal{M}$ $(n \in \mathbb{N})$ be pairwise disjoint measurable sets. If for all $n \in \mathbb{N}$, $|E_n| \leq |\mathbb{N}|$, then

$\mu\left(\bigcup_{n=1}^{\infty} E_n\right) = 0$, since countable union of countable sets is countable,

and clearly $\sum_{n=1}^{\infty} \mu(E_n) = 0$, since $\mu(E_n) = 0$ for all $n \in \mathbb{N}$.

otherwise, assume that for some $k \in \mathbb{N}$, $|E_k| > |\mathbb{N}|$.

Then, $\mu\left(\bigcup_{n=1}^{\infty} E_n\right) = 1$. Meanwhile, suppose that there is $r \neq k$ such that $|E_r| > |\mathbb{N}|$. Then, since $E_r, E_k \in \mathcal{M}$, $|E_r^c|, |E_k^c| \leq |\mathbb{N}|$.

And, since their are disjoint, $E_r^c \cup E_k^c = X$, but it contradicts, since X is uncountable. Finally, $\sum_{n=1}^{\infty} \mu(E_n) = 1$. so, μ is a measure.

$f^{-1}[(a, b)]$